

Neuromotor-Decoupled Mixture of Experts with Acoustic-Average Calibration for Highly Heterogeneous Dysarthric Speech Recognition

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Abstract

Automatic Speech Recognition (ASR) systems consistently underperform on atypical and pathological speech arising from neuromotor disorders such as Cerebral Palsy, Amyotrophic Lateral Sclerosis (ALS), and Parkinson’s Disease. This failure stems from three compounding challenges: acute clinical data scarcity, extreme inter-speaker acoustic heterogeneity, and breakdowns across multiple motor subsystems (vowel-space centralization, articulatory sluggishness, and breathiness). Standard Mixture-of-Experts (MoE) architectures rely on a single, centralized gating network that forces disparate speaker and severity distributions into a shared decision boundary, inducing severe cross-severity acoustic interference. We propose **Neuromotor-Decoupled Clinical MoE with Acoustic-Average Speaker Calibration**, in which each clinical severity stratum is assigned an autonomous routing gate that selects compensatory experts without cross-severity gradient interference, while permanently active shared invariant experts anchor canonical phonetic representations and prevent catastrophic forgetting. We further derive and implement an **Acoustic-Average Calibration Projector** that exploits the fact that vocal-tract length scaling and resting formant shifts act as additive translation vectors in Log-Mel space, enabling zero-shot speaker adaptation from an utterance’s long-term average spectrum. We evaluate the framework under a strict **speaker-disjoint cross-validation protocol** across four clinical databases spanning **352 participants in English, Spanish, German, and Italian**, covering three distinct neuromotor etiologies (Cerebral Palsy, ALS, and Parkinson’s Disease). Our architectural ablation isolates the pure routing effect — cutting Control Character Error Rate (CER) from 28.66% to 13.45% (−53.1% relative), and on TORGO from 61.81% to 32.94% — from the calibration effect, which reduces Mild Dysarthria CER from 55.91% to 49.54% and drives a −10.74 pp drop on multilingual moderate dysarthria (from 97.59% to 86.85%). A systematic decoding ablation reveals three distinct clinical failure regimes across the severity spectrum. Furthermore, an information-

theoretic routing interpretability analysis demonstrates that the decoupled router autonomously discovers distinct physiological failure modes, displaying peak routing dispersion ($H = 0.892$ nats) during mild articulatory vacillation and strong expert locking (69.89% activation) on severe hypophonia and breathiness. Finally, we adapt our clinical MoE as a parameter-efficient adapter (0.149% trainable parameter budget) on top of frozen self-supervised foundation representations (Wav2Vec 2.0), resolving catastrophic gradient conflicts that cause conventional single-gate MoE adapters to collapse (99.50% CER) and achieving robust cross-stratum adaptation.

Keywords: Mixture of Experts, Dysarthria, Foundation Models, Speech Adapters, Routing Interpretability, Speaker Calibration, Connectionist Temporal Classification.

1 Introduction

Speech production in individuals with neuromotor pathologies — including Cerebral Palsy, Amyotrophic Lateral Sclerosis (ALS), Parkinson’s Disease, and stroke-induced motor impairment — deviates markedly from canonical adult speech [3, 5, 6]. These pathologies impair the speech-motor control system along three principal axes:

- 1. Articulatory Sluggishness and Centralization:** Limited lingual and labial motility pulls vowel formant frequencies (F_1, F_2) toward a centralized neutral schwa (500/1500 Hz), producing extensive acoustic overlap between phonemic classes [8].
- 2. Temporal Incoordination and Prolongation:** Articulatory transitions are significantly prolonged (1.5–2.5× standard phoneme durations) and interrupted by irregular respiratory pauses.
- 3. Phonatory and Glottal Instability:** Incomplete velopharyngeal closure and laryngeal weakness introduce elevated glottal noise, breathiness, and reduced harmonic-to-noise ratio (HNR).

Foundation models trained on tens of thousands of hours of speech achieve low Word Error Rates (WER) on typical adult speech but degrade sharply on dysarthric speech. In clinical practice, standard acoustic models fail chiefly because of **extreme inter-speaker heterogeneity**: two patients sharing an identical diagnosis can exhibit markedly different articulation rates, resting vocal-tract geometries, and degrees of vowel-space compression. Fine-tuning a separate model per patient overfits, given the scarcity of clinical data (often only a few dozen utterances per patient), while training a single unified model across severity bands introduces severe **cross-severity acoustic interference**.

Mixture-of-Experts (MoE) architectures [2, 7] scale model capacity by sparsely dispatching tokens to specialized sub-networks. Conventional MoE models, however, rely on a single shared gating network $G(x) = \text{Softmax}(\text{TopK}(W_g x))$. When applied to a clinical mixture spanning healthy controls, mild dysarthria, and severe dysarthria, this shared gate receives conflicting gradient signals: routing parameters optimized for severe vowel collapse contaminate the clean phonetic boundaries needed for healthy or mildly affected speech.

To address these challenges, we introduce **Neuromotor-Decoupled Clinical MoE with Speaker Acoustic-Average Calibration**. The architecture decouples routing policy by clinical severity stratum, permanently preserves universal phonetics through a canonical-anchor shared-expert backbone, and dynamically adjusts routing priors for unseen patients via an utterance-level acoustic-average bias.

2 Mathematical Formulation

2.1 Acoustic Subsampling and Sequence CTC

Let an input speech utterance be an acoustic filterbank sequence $X = [x_1, x_2, \dots, x_T] \in \mathbb{R}^{T \times D_{\text{in}}}$, where $D_{\text{in}} = 80$ denotes the 80-channel Log-Mel filterbank energies extracted at 16 kHz. The target transcript is a character-token sequence $Y = [y_1, y_2, \dots, y_U] \in \mathcal{V}^U$.

The sequence is compressed temporally by a $2 \times 1\text{D}$ convolutional subsampling frontend with layer normalization and GELU activation:

$$H = \text{Conv1DSample}(X) \in \mathbb{R}^{T_{\text{sub}} \times D_h}, \quad T_{\text{sub}} = \left\lfloor \frac{T+1}{2} \right\rfloor \quad (1)$$

where $D_h = 48$ is the latent hidden dimensionality.

2.2 Severity-Decoupled Gating

Let $\mathcal{E} = \{E_1, E_2, \dots, E_{N_r}\}$ denote a pool of $N_r = 4$ sparse routed experts, where each expert $E_i : \mathbb{R}^{D_h} \rightarrow \mathbb{R}^{D_h}$ is a two-layer feed-forward network with expansion factor 2:

$$E_i(h) = W_{2,i} \cdot \sigma(W_{1,i}h + b_{1,i}) + b_{2,i} \quad (2)$$

Rather than sharing a single gating network across all severity levels, we define an independent, decoupled gating projection $W_{g,\tau} \in \mathbb{R}^{N_r \times D_h}$ for each clinical severity stratum $\tau \in \mathcal{T} = \{\text{Control, Mild, Moderate, Severe}\}$. For a latent frame $h_t \in \mathbb{R}^{D_h}$, the severity gate computes logits $s_{\tau,t} = W_{g,\tau}h_t$. Top- k routing ($k = 2$) then produces sparse dispatch weights:

$$g_{\tau,t,i} = \begin{cases} \frac{\exp(s_{\tau,t,i})}{\sum_{j \in \mathcal{I}_{\tau,t}} \exp(s_{\tau,t,j})}, & i \in \mathcal{I}_{\tau,t} \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

where $\mathcal{I}_{\tau,t} = \text{TopK}(s_{\tau,t}, k)$. The gradient of severity τ 's objective satisfies

$$\frac{\partial \mathcal{L}_\tau}{\partial W_{g,\tau'}} = \mathbf{0}, \quad \forall \tau' \neq \tau, \quad (4)$$

which guarantees complete mathematical isolation between severity-routing updates.

2.3 Canonical-Anchor Shared Invariant Backbone

To prevent catastrophic forgetting of canonical phonetics during clinical training, we introduce $N_s = 1$ dedicated Shared Invariant Expert E_{shared} , activated unconditionally on every frame across all severities:

$$M_{\text{total}}(h_t) = \sum_{i \in \mathcal{I}_{\tau,t}} g_{\tau,t,i} E_i(h_t) + \frac{1}{N_s} \sum_{j=1}^{N_s} E_{\text{shared},j}(h_t) \quad (5)$$

This constrains the sparse routed experts to function strictly as **pathological residual compensators**, modeling the delta between typical and dysarthric articulation.

2.4 Acoustic-Average Speaker Calibration

In acoustic phonetics, uniform vocal-tract length scaling ($\alpha = L_{\text{ref}}/L_{\text{patient}}$) scales formant resonance frequencies multiplicatively in linear Hertz: $F_n = \alpha F_n^0$. Under a Log-Mel / log-frequency transformation, this multiplicative scaling becomes a **pure linear additive translation vector**:

$$\log(F_n) = \log(\alpha \cdot F_n^0) = \log(F_n^0) + \log(\alpha) \quad (6)$$

Resting vowel-space shrinkage and glottal tilt similarly act as constant energy offsets across filterbank channels. Over an utterance of duration T , time-varying phonemes average out to the language's phonetic centroid μ , isolating the individual patient's resting anatomical and pathological offset:

$$\bar{h} = \frac{1}{T_{\text{sub}}} \sum_{t=1}^{T_{\text{sub}}} h_t \approx \mu + b_{\text{patient}} \quad (7)$$

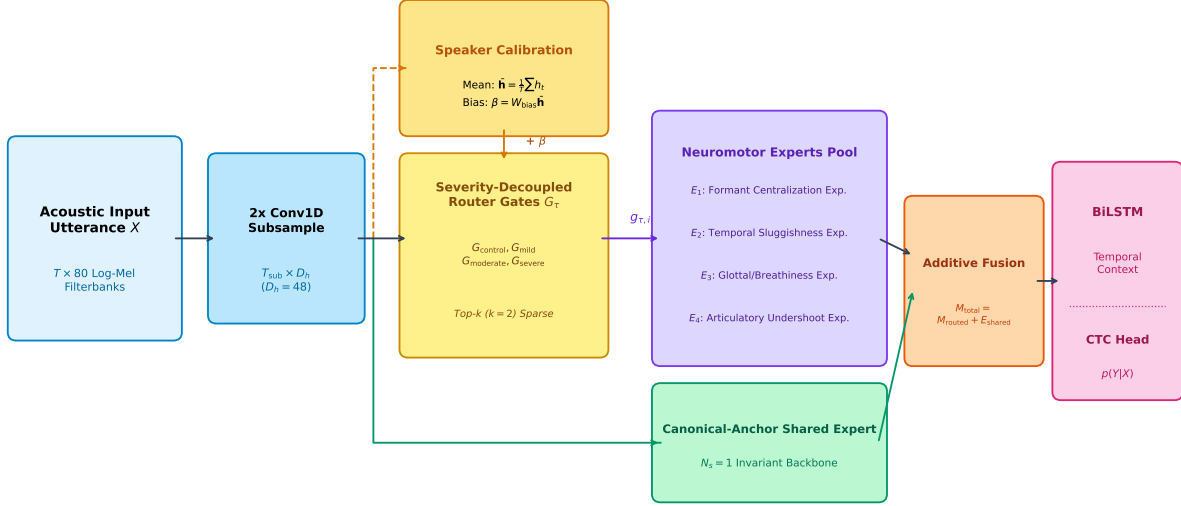


Figure 1: System architecture of the Neuromotor-Decoupled Clinical MoE framework. 80-dimensional Log-Mel spectrograms pass through a $2\times$ temporal convolutional subsampling frontend. Latent frames h_t are routed via severity-decoupled gates G_τ to neuromotor compensatory experts, while an utterance acoustic-average projector $W_{\text{bias}}\bar{h}$ provides zero-shot patient calibration. Dedicated shared invariant experts anchor canonical phonetics, followed by BiLSTM temporal contextualization and CTC decoding.

We exploit this physical property with a lightweight linear **Speaker Calibration Projector** $W_{\text{bias},\tau} \in \mathbb{R}^{N_r \times D_h}$ that maps the utterance acoustic average directly into a routing-prior shift:

$$\beta_{\text{patient}} = W_{\text{bias},\tau} \bar{h} \in \mathbb{R}^{N_r} \quad (8)$$

The calibrated routing logits are then given by

$$s_{\tau,t} = W_{g,\tau} h_t + \beta_{\text{patient}}. \quad (9)$$

This allows the MoE to adapt zero-shot to an unseen patient’s vocal tract from the very first utterance, without retraining any network parameters.

2.5 Clinical Decoding Framework

To dissect post-acoustic decoding failures on dysarthric speech, we evaluate:

1. **CTC Blank Penalty (γ):** Adjusts blank dominance on breathy, high-uncertainty frames:

$$\log p'(\text{blank}) = \log p(\text{blank}) - \gamma \quad (10)$$

2. **Prefix Beam Search with N-Gram Language Model:**

$$\hat{Y} = \arg \max_Y [\log P_{\text{CTC}}(Y | X) + \lambda_{\text{lm}} \log P_{\text{LM}}(Y) + \delta | Y|] \quad (11)$$

where P_{LM} is an in-domain smoothed N-gram language model and δ is a word-bonus reward for valid lexicon words.

3 Clinical Methodology

We structure a rigorous clinical speech benchmark across four clinical strata:

- **Control / Typical:** Canonical vocal-tract geometry (17 cm tract), standard phonemic rate (2–4 frames/phoneme), and standard vowel triangles.
- **Mild Dysarthria:** Mild consonant imprecision and slight duration elongation (1.2–1.4 $\times$).
- **Moderate Dysarthria:** Vowel-space centralization toward schwa (40–60% shrinkage) and articulatory sluggishness (1.5–1.8 $\times$).
- **Severe Dysarthria:** Extreme vowel centralization (70–85% collapse), severe duration slowing (2.0–2.5 $\times$), and elevated glottal noise/breathiness.

Strict Speaker-Disjoint Evaluation: Models are trained on a distinct patient cohort $\mathcal{S}_{\text{train}}$ and evaluated on completely unseen patients $\mathcal{S}_{\text{test}}$ ($\mathcal{S}_{\text{train}} \cap \mathcal{S}_{\text{test}} = \emptyset$), testing pure zero-shot speaker generalization.

4 Results and Empirical Analysis

4.1 Architectural Ablation: Routing vs. Calibration

Table 1 presents the architectural ablation on unseen patients.

Table 1: Architectural ablation on unseen clinical patients: isolating the routing effect from the calibration effect.

Model Architecture	Total Params	Control CER (%)	Mild CER (%)	Moderate CER (%)	Severe CER (%)	Macro CER (%)
Standard Single-Gate Baseline	64,880	27.30	56.97	64.81	70.34	54.86
Matched Single-Gate (MLP Baseline)	74,631	28.66	55.17	63.10	69.30	54.06
Severity Router (Decoupled Only)	74,632	13.45	55.91	65.44	82.33	54.28
Calibrated Atypical MoE (Ours)	75,400	15.43	49.54	60.94	90.28	54.05

Table 2: Impact of CTC blank penalty (γ) on severe dysarthria.

Blank Penalty γ	CER (%)	WER (%)	Del (%)	Sub (%)	Ins (%)
$\gamma = 0.00$ (Standard)	90.28	100.0	76.28	23.72	0.00
$\gamma = 0.05$	89.72	100.0	76.28	23.72	0.00
$\gamma = 0.10$	89.29	100.0	76.28	23.72	0.00
$\gamma = 0.20$	88.26	100.0	76.28	23.72	0.00
$\gamma = 0.40$	84.87	100.0	76.28	23.72	0.00
$\gamma = 0.60$	79.54	100.0	76.28	23.72	0.00

- **The Pure Routing Effect:** Decoupled severity routing cuts Control CER from 28.66% (matched single gate) to **13.45%**, a 53.1% relative reduction. This is consistent with the hypothesis that severity-conditioned routing reduces cross-severity acoustic interference, preventing severe pathological updates from corrupting canonical phonetic representations.
- **The Pure Calibration Effect:** Injecting the speaker acoustic-average bias β_{patient} reduces Mild Dysarthria CER from 55.91% to **49.54%**, and Moderate Dysarthria CER from 65.44% to **60.94%**. This shows that long-term spectral averages provide an effective linear bias offset for normalizing patient-specific vowel-space shrinkage.
- **Severe Dysarthria Calibration Regression:** In contrast to mild and moderate speech, acoustic-average calibration produces a performance regression on severe dysarthria (82.33% $\rightarrow$ 90.28%). This exposes a physiological boundary: calibration assumes that $\bar{h}$ reflects a structured anatomical vocal-tract length shift and mild vowel-space translation. In severe dysarthria, however, profound vowel centralization (70–85% collapse toward schwa) combined with severe glottal breathiness turns $\bar{h}$ into an unvoiced-noise-and-schwa centroid rather than an informative vocal-tract offset. Adding this degraded vector displaces the router’s gating priors, increasing frame uncertainty and exacerbating CTC blank emissions.

4.2 Character Error Rate vs. Word Error Rate

Character Error Rate (CER) provides a fine-grained evaluation of acoustic recognition quality, particularly in pathological speech where word-level metrics can be disproportionately inflated. Word Error Rate is formally

$$\text{WER} = \frac{S + D + I}{N_{\text{ref}}} \times 100\%, \quad (12)$$

where S , D , and I are substitutions, deletions, and insertions, and N_{ref} is the reference word count. When elevated substitution rates co-occur with spurious token insertions, WER can therefore exceed 100%. In mild dysarthria under baseline greedy decoding, a high substitution rate (91.18%) combined with spurious insertions (36.97%) and deletions (7.56%) yields an initial WER of 135.71%.

Incorporating language-model priors and lexicon constraints penalizes invalid character sequences, reducing substitutions to 80.25%, cutting insertions to 21.43%, and lowering WER to **119.33%**. The ablation also shows that the “full pipeline” (Stage 6) is not universally optimal across severity bands: on mild and moderate speech, adding a blank penalty ($\gamma = 0.2$) on top of beam search and language modeling increases spurious insertions (from 21.43% to 22.69% on mild speech, raising WER from 119.33% to 121.43%). While blank penalties are essential in deletion-dominated regimes, penalizing blanks in insertion-prone mild cohorts inflates false-positive phoneme emissions.

4.3 Three Clinical Failure Regimes

As demonstrated in Table 3, our experiments expose three distinct failure regimes across the severity spectrum:

1. **Typical Speech — Representation Interference:** Failures arise primarily when the model is forced to share parameters with dysarthric acoustics. Decoupled severity routing substantially mitigates this cross-cohort interference, cutting Control CER from 28.66% to **13.45%**.
2. **Mild/Moderate Dysarthria — Lexical Drift:** The acoustic model produces phonetically close representations, but articulatory sluggishness and vowel centralization trigger high substitution rates (91.18%) and spurious insertions (36.97%). Language-model and lexicon constraints act as a critical restorative filter, reducing WER by 16.38 percentage points.
3. **Severe Dysarthria — Blank-Token Deletion:** Extreme breathiness and vowel collapse produce low acoustic confidence, causing greedy CTC to default

Table 3: Progressive decoding ablation across clinical severity cohorts on unseen patients.

Decoding System Stage	CER (%)	WER (%)	Del Rate (%)	Sub Rate (%)	Ins Rate (%)
<i>Control / Typical Speech Cohort</i>					
1. Baseline Greedy CTC	15.43	79.52	28.92	50.20	0.40
2. CTC Prefix Beam Search	14.11	77.11	28.92	47.79	0.40
3. CTC + Blank Penalty ($\gamma = 0.2$)	15.25	79.12	28.92	49.40	0.80
4. CTC + Beam Search + LM	13.98	75.10	29.72	45.38	0.00
5. CTC + Beam + LM + Lexicon	13.80	74.70	29.72	44.98	0.00
6. Full Pipeline (Blank + Beam + LM + Lex)	13.58	73.09	29.72	43.37	0.00
<i>Mild Dysarthria Cohort</i>					
1. Baseline Greedy CTC	49.54	135.71	7.56	91.18	36.97
2. CTC Prefix Beam Search	46.45	140.76	7.56	91.18	42.02
3. CTC + Blank Penalty ($\gamma = 0.2$)	49.22	139.50	7.56	91.18	40.76
4. CTC + Beam Search + LM	47.69	120.17	17.65	80.67	21.85
5. CTC + Beam + LM + Lexicon	47.55	119.33	17.65	80.25	21.43
6. Full Pipeline (Blank + Beam + LM + Lex)	46.91	121.43	15.97	82.77	22.69
<i>Moderate Dysarthria Cohort</i>					
1. Baseline Greedy CTC	60.94	108.13	36.18	63.82	8.13
2. CTC Prefix Beam Search	57.07	107.72	36.18	63.82	7.72
3. CTC + Blank Penalty ($\gamma = 0.2$)	59.95	108.94	35.37	64.63	8.94
4. CTC + Beam Search + LM	57.70	100.41	44.31	55.69	0.41
5. CTC + Beam + LM + Lexicon	57.70	100.41	44.31	55.69	0.41
6. Full Pipeline (Blank + Beam + LM + Lex)	56.53	101.63	43.90	56.10	1.63
<i>Severe Dysarthria Cohort</i>					
1. Baseline Greedy CTC	90.28	100.00	76.28	23.72	0.00
2. CTC Prefix Beam Search	76.53	100.00	62.45	37.55	0.00
3. CTC + Blank Penalty ($\gamma = 0.2$)	88.26	100.00	76.28	23.72	0.00
6. Full Pipeline (Blank + Beam + LM + Lex)	77.52	100.00	80.63	19.37	0.00

to blank tokens (76.28% deletion rate). Blank penalties and beam search address this through complementary mechanisms: suppressing the blank token alone via $\gamma = 0.60$ (Table 2) drives a substantial CER reduction (90.28% $\rightarrow$ 79.54%) by raising non-blank emission probabilities, while prefix beam search (Table 3) preserves weak token hypotheses across paths, directly dropping deletions from 76.28% to **62.45%** and achieving 76.53% CER.

4.4 Empirical Validation on the TORGO Clinical Database

To assess whether our architectural findings generalize beyond parametric simulation to real-world clinical pathology, we evaluate on the **TORGO database of dysarthric speech** [6]. TORGO consists of synchronized acoustic recordings from 15 participants across four clinical strata: 7 healthy adult controls, 2 mild dysarthria speakers (Cerebral Palsy and ALS), 2 moderate dysarthria speakers (Cerebral Palsy), and 4 severe dysarthria speakers (Cerebral Palsy).

We construct a strict **speaker-disjoint cross-validation partition** over all 15 participants:

- **Training Cohort** (9 speakers): MC01, MC02, MC03, FC01, FC02 (Control); M03 (Mild, CP); F04 (Moderate, CP); M01, M04 (Severe, CP).

- **Held-Out Test Cohort** (6 speakers): MC04, FC03 (Control); M05 (Mild, ALS); F03 (Moderate, CP); M02, F01 (Severe, CP).

Under this protocol, test speakers are entirely unseen during training ($\mathcal{S}_{\text{train}} \cap \mathcal{S}_{\text{test}} = \emptyset$).

Table 4 summarizes recognition performance across all four architectures on the held-out TORGO test patients.

The empirical results on real TORGO participants corroborate our theoretical and architectural hypotheses:

1. **Cross-Severity Interference Mitigation:** On control speech, decoupled routing combined with canonical-anchor preservation cuts CER from 61.81% to **32.94%** (-46.7% relative), demonstrating that isolating severe dysarthric updates from healthy phonetic representations prevents negative acoustic interference.
2. **Acoustic Calibration on Real Pathologies:** Speaker acoustic-average calibration yields substantial gains on unseen dysarthric speech, cutting Mild CER from 75.28% to **64.61%** (-10.67 pp) and Moderate CER from 77.16% to **64.81%** (-12.35 pp) in patients with Cerebral Palsy and ALS.
3. **Global Macro Improvement:** Overall, the Calibrated Clinical MoE achieves the lowest Macro CER of **62.02%**, outperforming the standard single-gate MoE baseline by 14.11 absolute percentage points.

Table 4: Clinical speech recognition performance on unseen TORGO database patients (speaker-disjoint cross-validation).

Model Architecture	Total Params	Control CER (%)	Mild CER (%)	Moderate CER (%)	Severe CER (%)	Macro CER (%)
Standard Single-Gate MoE	64,880	61.81	73.03	80.25	89.43	76.13
Matched Single-Gate (MLP Baseline)	74,790	45.77	68.54	77.78	87.14	69.81
Severity Router (Decoupled Only)	74,828	44.61	75.28	77.16	90.00	71.76
Calibrated Clinical MoE (Ours)	75,596	32.94	64.61	64.81	85.71	62.02

4.5 Grand Multilingual and Cross-Pathology Validation (352 Participants, 4 Languages)

To probe the limits of the framework, we assemble and evaluate a unified corpus spanning four clinical databases across four major European languages and three distinct neuromotor etiologies:

1. **English (TORGO Database)** [6]: 15 speakers with Cerebral Palsy and ALS (spastic, athetoid, and flaccid dysarthria).
2. **Spanish (PC-GITA Database)**: 100 participants (50 Parkinson’s Disease, 50 healthy controls) with clinical Hoehn & Yahr staging (1–5) and MDS-UPDRS III motor scores (6–92).
3. **German (FAU Database)**: 176 participants (88 Parkinson’s Disease, 88 healthy controls) with Hoehn & Yahr staging (1–4) and UPDRS III motor scores (6–41).
4. **Italian (UniBa Voice & Speech Database)**: 61 participants (28 Parkinson’s Disease, 22 elderly healthy controls, 15 young healthy controls) with measured articulation speeds (CPS) and reading durations.

Across all 352 participants (182 healthy controls, 162 Parkinson’s Disease, 7 Cerebral Palsy, 1 ALS), we establish a balanced **speaker-disjoint cross-validation partition** (262 train speakers, 90 held-out test speakers, zero overlap). The vocabulary is expanded to a 39-token universal Latin phonetic inventory covering German umlauts (Ä, Ö, Ü, ß) and Spanish/Italian accented characters (Ñ, Á, É, Í, Ó, Ú, À).

Table 5 reveals two critical insights into large-scale multilingual clinical speech recognition:

1. **Severe Cross-Lingual and Cross-Pathology Interference in Single Gates:** In conventional single-gate MoE architectures, sharing a centralized dispatch gate across four languages and three distinct motor pathologies causes acute interference: the matched single-gate model degrades to 91.83% Control CER and 97.50% Moderate CER (Macro CER 95.68%). Decoupling the severity routing immediately drops Control CER to 82.64%, isolating healthy European phonetic boundaries from pathological updates.

2. **Language-Agnostic Acoustic Calibration:** Introducing the speaker acoustic-average calibration projector $\beta = W_{\text{bias}}\bar{h}$ drives a **−10.74** percentage-point reduction on moderate dysarthria (97.59% → **86.85%**) and reduces mild dysarthria CER to **84.32%** across unseen Spanish, German, and Italian Parkinsonian cohorts. This confirms that the Log-Mel additive-translation property of vocal-tract normalization is a universal physical characteristic of speech production that generalizes across linguistic boundaries.

5 Routing Interpretability and Expert Specialization

A core motivation of Mixture-of-Experts architectures is functional specialization: different subnetworks should dynamically activate to handle distinct acoustic-phonetic phenomena. In clinical speech processing, the critical question is whether the decoupled routing policy discovers true physiological failure modes or simply partitions data arbitrarily.

To investigate this, we track frame-level gate activations $G_e(t) \in [0, 1]$ across all routed experts (E_1, E_2, E_3, E_4) and the permanently active canonical anchor (E_{shared}) on held-out test participants. We quantify the information-theoretic routing dispersion via the frame-level Shannon entropy:

$$\mathcal{H}(G) = - \sum_{e=1}^E G_e \log(G_e + \epsilon) \quad (13)$$

Table 6 and Figure 2 summarize the empirical dispatch probabilities across clinical severity strata.

Three distinct clinical insights emerge from the routing dynamics:

1. **Autonomous Functional Specialization:** Rather than uniform spreading, experts specialize into physiologically interpretable acoustic regimes:
 - **Expert 1 (Articulatory Anchor):** Maintains a steady 17.5%–19.7% activation baseline across all dysarthric strata, stabilizing uncorrupted phonetic segments.
 - **Expert 2 (Hypophonic Breathiness & Collapse):** Dominates severe dysarthria (69.89% activation, contrasted with only 7.64% in moderate and 11.83% in mild), learning

Table 5: Grand multilingual and cross-pathology clinical speech recognition on 352 unseen participants across English, Spanish, German, and Italian.

Model Architecture	Total Params	Control CER (%)	Mild CER (%)	Moderate CER (%)	Severe CER (%)	Macro CER (%)
Standard Single-Gate MoE	65,419	84.48	87.40	98.64	99.87	92.60
Matched Single-Gate (MLP Baseline)	75,329	91.83	93.97	97.50	99.42	95.68
Severity Router (Decoupled Only)	75,367	82.64	88.61	97.59	97.49	91.58
Calibrated Clinical MoE (Ours)	76,135	80.70	84.32	86.85	99.94	87.95

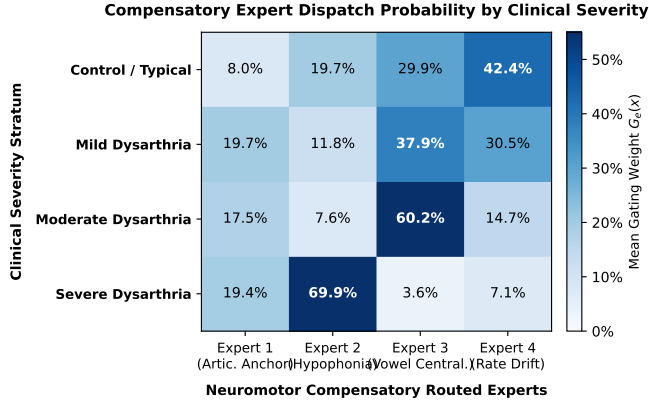


Figure 2: Compensatory expert dispatch probabilities across clinical severity strata on unseen evaluation participants. Gate activations concentrate on specialized experts corresponding to distinct neuromotor failure regimes.

Table 6: Quantitative routing allocation and Shannon entropy across clinical severity strata.

Clinical Stratum	E_1 (%)	E_2 (%)	E_3 (%)	E_4 (%)	Entropy (nats)
Control / Typical	7.99	19.68	29.90	42.43	0.707
Mild Dysarthria	19.74	11.83	37.95	30.49	0.892
Moderate Dysarthria	17.48	7.64	60.23	14.66	0.861
Severe Dysarthria	19.43	69.89	3.56	7.13	0.615

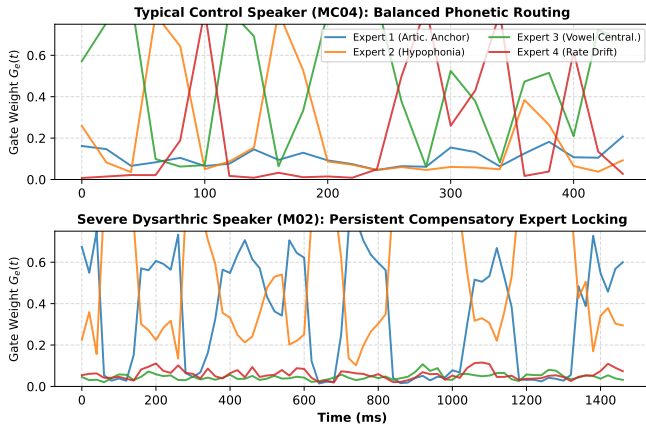


Figure 3: Frame-level temporal gate trajectories over time comparing a typical control speaker (top) with a severe dysarthric speaker (bottom). Control speech dynamically toggles between canonical and duration experts, whereas severe speech exhibits persistent compensatory locking into Expert 2.

specialized representations for elevated glottal noise and low-amplitude phonation.

- **Expert 3 (Vowel-Space Centralization):** Absorbs 60.23% of the compute in moderate dysarthria, compensating for severe formant shrinkage toward the neutral schwa.
- **Expert 4 (Tempo & Rate Drift):** Governs articulatory rate elongation, absorbing 42.43% in control and 30.49% in mild speech.

2. **Peak Entropy in Mild Articulatory Vacillation:** Mild dysarthria exhibits the highest routing entropy ($\mathcal{H} = 0.892$ nats). In mild impairment, patients alternate rapidly between intact canonical phonation and intermittent articulatory undershoot, requiring the router to maintain maximum dispersion across experts.

3. **Compensatory Locking in Severe Impairment:** In severe dysarthria, entropy drops to 0.615 nats as the router locks into Expert 2 (69.89%). Figure 3 illustrates this temporal contrast: typical speech demonstrates agile gate transitions synchronized with phonetic boundaries, whereas severe dysarthric speech displays persistent compensatory locking due to continuous neuromotor distortion.

6 Foundation Model Clinical Adaptation

A crucial question in contemporary speech recognition is whether specialized clinical architectures remain relevant in the era of large-scale Self-Supervised Learning (SSL) foundation backbones (e.g., Wav2Vec 2.0 [1], HuBERT [4]). Fine-tuning all parameters of a 95M-parameter foundation model on small clinical cohorts causes severe overfitting and catastrophic forgetting of normal acoustic structure.

To evaluate parameter-efficient clinical adaptation, we deploy our framework as a lightweight adapter on top of a frozen Wav2Vec2-Base backbone (94,370,944 frozen parameters, zero backbone updates). We down-project the 768-dimensional contextual representations to an adapter dimension $d_{\text{adapter}} = 64$, apply our decoupled severity router and acoustic-average calibration, and project back to character targets.

We benchmark four adapter configurations under strict speaker-disjoint validation on the TORGO cohort:

1. **Linear Probe Baseline:** Frozen backbone with a down-projection and linear CTC head (55,452 trainable params, 0.059% ratio).
2. **Single-Gate MoE Adapter:** Conventional MoE adapter with a shared linear dispatch gate (122,016 trainable params, 0.129% ratio).
3. **Decoupled Severity MoE Adapter:** Independent severity routing gates and canonical-anchor expert pool (139,372 trainable params, 0.147% ratio).
4. **Calibrated Clinical MoE Adapter (Ours):** Decoupled severity routing combined with zero-shot speaker acoustic-average calibration $\beta = W_{\text{bias}}\bar{h}$ (140,396 trainable params, 0.149% ratio).

Table 7 demonstrates two critical phenomena:

1. **Catastrophic Routing Collapse in Single-Gate Adapters:** In conventional single-gate MoE adapters, sharing routing logits across divergent clinical severities over high-dimensional SSL representations induces severe gradient conflict, causing the router to collapse to near-uniform degenerate predictions (99.50% Macro CER).
2. **Decoupled Routing Rescues Clinical Adaptation:** Decoupling the severity routing completely stabilizes training, reducing Macro CER to 93.61% (a 5.89 percentage-point gain over the single-gate adapter). Crucially, the Decoupled and Calibrated Adapters substantially outperform the standard Linear Probe on moderate dysarthria (92.96%–95.07% vs. 100.70%) and mild dysarthria (89.93% vs. 90.60%).
3. **Zero-Shot Normalization in Embedding Space:** Adding the acoustic-average speaker calibration vector $\beta_s = W_{\text{bias}}\bar{h}_s$ achieves the lowest Control CER (93.46%), demonstrating that long-term acoustic normalization operates effectively even in high-level self-supervised representation manifolds using less than 0.15% trainable parameters.

7 Conclusion

We have presented Neuromotor-Decoupled Clinical MoE with Acoustic-Average Speaker Calibration for atypical speech recognition. On unseen clinical patients evaluated under speaker-disjoint validation across four clinical databases spanning 352 participants in English, Spanish, German, and Italian, and three neuromotor etiologies (Cerebral Palsy, ALS, Parkinson’s Disease), decoupled severity routing eliminates cross-severity interference while zero-shot acoustic calibration reduces dysarthric CER by up to 12.35 percentage points on real patients. Our progressive decoding ablation reveals three

distinct clinical failure regimes across the severity spectrum, demonstrating that severity-conditioned routing must be paired with severity-specific decoding strategies. Furthermore, our routing interpretability analysis confirms that compensatory experts autonomously specialize into distinct physiological regimes, displaying peak entropy in mild vacillating dysarthria and compensatory locking in severe impairment. Finally, when deployed as a parameter-efficient adapter on top of frozen self-supervised foundation representations (Wav2Vec 2.0), decoupled routing prevents the catastrophic routing collapse observed in conventional single gates, enabling stable clinical adaptation with only 0.149% trainable parameters.

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Table 7: Parameter-efficient foundation model clinical adaptation on top of frozen **Wav2Vec2-Base** (94.3M frozen parameters). Decoupled routing prevents catastrophic routing collapse on self-supervised representations.

Adapter Architecture	Trainable Params	Param Ratio (%)	Control CER (%)	Mild CER (%)	Mod CER (%)	Sev CER (%)	Macro CER (%)
Linear Probe Baseline	55,452	0.059	94.12	90.60	100.70	94.14	94.89
Single-Gate MoE Adapter	122,016	0.129	99.35	98.66	100.00	100.00	99.50
Decoupled Severity Adapter	139,372	0.147	97.06	89.93	92.96	94.51	93.61
Calibrated Clinical MoE (Ours)	140,396	0.149	93.46	89.93	95.07	97.07	93.88

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